Research Evidence Pack — Read First

Review scope	v29 matrix: 21 entries (17 targeted additions + 4 carry-forward anchors).
Current code package	v26 iFogSim2; 192 Python tests reported passing; 448 Java source files compiled; two pre-existing deprecation warnings.
Training/evaluation	3 training seeds 801–803; 81-state Q table; v26 60 runs (2 scenarios × 6 named policies × 5 test seeds).
Older experiment set	v25 summary: 4 scenarios × 5 policy mechanisms × 5 seeds = 100 run combinations.
Legacy technical audit	Seed 1/2; 228 generated tasks per run; cutoff 500; old fixed-deadline setting.
Policy naming	gamma=0 and decision_model=contextual_bandit in current v26 config.
Main performance result	Q policy wins mean latency on v25 edge capacity ×0.5; loses to latency-greedy on base/higher load and both new v26 scenarios; equal to greedy on network ×2.
Research gap	No broad algorithmic gap confirmed in the current focused literature audit.
Metrics caveats	Latency ends at first processing-module completion; utilization and traffic are proxies; energy not task-level logged.
Next action	Freeze one version/config/model, close literature evidence gaps, agree metric definitions, run final independent seeds/topologies.

Not a systematic review; native database exports and full screening counts not complete.

The report uses supplied logs/README; no full rerun performed during synthesis.

deadline_greedy duplicates latency_greedy; effectively five unique policies.

Separate from v26; do not pool versions without a protocol/version audit.

History only; not the same experiment as v26 heterogeneous deadlines.

Call it tabular contextual-bandit/Q-value policy unless sequential Q-learning is reimplemented with tested gamma>0.

No claim of general superiority.

Supervisor must choose new algorithm vs comparative/reproducibility study.

No full application-loop, CPU-utilization, total per-hop traffic, or energy-benefit claims yet.

Do not tune or choose the method based on final test seeds.

Archived v25 results — keep separate from v26

Scenario	Policy	Seeds
Base / baseline workload	Tabular Q-value / contextual-bandit policy	5
Base / baseline workload	Latency-greedy heuristic	5
Base / baseline workload	Edge-only	5
Base / baseline workload	Local-only	5
Base / baseline workload	Cloud-only	5
Edge capacity 50%	Tabular Q-value / contextual-bandit policy	5
Edge capacity 50%	Latency-greedy heuristic	5
Edge capacity 50%	Edge-only	5
Edge capacity 50%	Local-only	5
Edge capacity 50%	Cloud-only	5
Higher offered load (arrival interval ×0.7)	Tabular Q-value / contextual-bandit policy	5
Higher offered load (arrival interval ×0.7)	Latency-greedy heuristic	5
Higher offered load (arrival interval ×0.7)	Edge-only	5
Higher offered load (arrival interval ×0.7)	Local-only	5
Higher offered load (arrival interval ×0.7)	Cloud-only	5
Network propagation delay ×2	Tabular Q-value / contextual-bandit policy	5
Network propagation delay ×2	Latency-greedy heuristic	5
Network propagation delay ×2	Edge-only	5
Network propagation delay ×2	Local-only	5
Network propagation delay ×2	Cloud-only	5

Tasks across 5 seeds	Mean latency	Mean CI95 low	Mean CI95 high	P95 latency	P99 latency
6491	10.663	10.581	10.744	24.380	36.849
6491	8.275	8.185	8.365	24.333	37.011
6491	14.077	13.954	14.201	40.893	58.080
6491	20.314	19.729	20.899	98.286	123.076
6491	87.201	86.801	87.601	130.188	133.132
6519	11.654	11.453	11.856	37.667	52.019
6519	13.994	13.548	14.441	37.667	102.704
6519	86.061	73.654	98.468	337.798	708.090
6519	20.654	19.847	21.461	98.506	123.779
6519	87.370	86.916	87.824	130.371	133.179
9255	11.450	11.308	11.593	25.525	45.354
9255	9.553	9.396	9.710	24.398	39.581
9255	15.269	15.022	15.515	50.335	67.011
9255	126.100	118.478	133.721	513.166	1118.606
9255	87.653	87.398	87.908	130.266	133.110
6472	11.083	10.872	11.294	29.333	43.073
6472	11.083	10.872	11.294	29.333	43.073
6472	19.137	19.005	19.268	46.665	62.683
6472	20.130	19.684	20.576	95.899	120.445
6472	171.832	171.283	172.382	255.482	258.679

Deadline violation rate	Completion rate
0.000%	100.000%
0.000%	100.000%
0.000%	100.000%
5.962%	100.000%
49.900%	100.000%
0.000%	100.000%
1.902%	100.000%
21.933%	100.000%
6.075%	100.000%
50.086%	100.000%
0.000%	100.000%
0.000%	100.000%
0.022%	100.000%
27.087%	100.000%
50.265%	100.000%
0.000%	100.000%
0.000%	100.000%
0.000%	100.000%
5.856%	100.000%
100.000%	100.000%

Current v26 results — 2 scenarios, 5 seeds, 6 policy labels

Scenario	Policy	Seeds
Heterogeneous task deadlines	Tabular Q-value / contextual-bandit policy	5
Heterogeneous task deadlines	Latency-greedy heuristic	5
Heterogeneous task deadlines	Deadline-greedy diagnostic (duplicate)	5
Heterogeneous task deadlines	Edge-only	5
Heterogeneous task deadlines	Local-only	5
Heterogeneous task deadlines	Cloud-only	5
Edge-1 compute-service outage/recovery	Tabular Q-value / contextual-bandit policy	5
Edge-1 compute-service outage/recovery	Latency-greedy heuristic	5
Edge-1 compute-service outage/recovery	Deadline-greedy diagnostic (duplicate)	5
Edge-1 compute-service outage/recovery	Edge-only	5
Edge-1 compute-service outage/recovery	Local-only	5
Edge-1 compute-service outage/recovery	Cloud-only	5

Tasks across 5 seeds	Mean latency	Mean CI95 low	Mean CI95 high	Mean latency SD	P95 latency
1427	10.894	10.179	11.610	0.577	26.013
1427	8.158	7.661	8.654	0.400	24.333
1427	8.158	7.661	8.654	0.400	24.333
1427	14.412	13.643	15.181	0.619	40.583
1427	18.985	18.351	19.620	0.511	75.228
1427	87.166	86.469	87.862	0.561	130.077
1441	12.983	11.907	14.059	0.866	34.198
1441	11.272	10.255	12.290	0.820	33.966
1441	11.272	10.255	12.290	0.820	33.966
1441	15.250	14.526	15.973	0.583	40.659
1441	18.846	17.784	19.908	0.855	76.346
1441	87.029	85.825	88.234	0.970	129.969

P99 latency	Deadline violation rate	Deadline CI95 low	Deadline CI95 high	Completion rate
74.819	0.000%	0.000%	0.000%	100.000%
41.007	0.000%	0.000%	0.000%	100.000%
41.007	0.000%	0.000%	0.000%	100.000%
62.236	0.137%	-0.243%	0.517%	100.000%
119.215	1.403%	0.296%	2.509%	100.000%
133.334	88.092%	87.664%	88.520%	100.000%
106.198	0.209%	-0.181%	0.598%	100.000%
89.356	0.141%	-0.250%	0.532%	100.000%
89.356	0.141%	-0.250%	0.532%	100.000%
84.650	0.070%	-0.125%	0.266%	100.000%
116.709	0.765%	0.402%	1.129%	100.000%
133.113	88.263%	87.004%	89.521%	100.000%

Small miss rate	Medium miss rate	Large miss rate
0.000%	0.000%	0.000%
0.000%	0.000%	0.000%
0.000%	0.000%	0.000%
0.183%	0.000%	0.000%
0.000%	1.132%	21.111%
100.000%	51.666%	50.000%
0.000%	0.000%	3.750%
0.000%	0.000%	2.500%
0.000%	0.000%	2.500%
0.000%	0.000%	1.250%
0.000%	0.357%	12.710%
100.000%	52.850%	50.588%

v26 paired differences — exploratory, n=5 seeds

Scenario	Comparison
Heterogeneous task deadlines	qlearning minus latency_greedy
Heterogeneous task deadlines	qlearning minus latency_greedy
Heterogeneous task deadlines	qlearning minus latency_greedy
Heterogeneous task deadlines	qlearning minus latency_greedy
Heterogeneous task deadlines	qlearning minus deadline_greedy
Heterogeneous task deadlines	qlearning minus deadline_greedy
Heterogeneous task deadlines	qlearning minus deadline_greedy
Heterogeneous task deadlines	qlearning minus deadline_greedy
Heterogeneous task deadlines	qlearning minus edge
Heterogeneous task deadlines	qlearning minus edge
Heterogeneous task deadlines	qlearning minus edge
Heterogeneous task deadlines	qlearning minus edge
Heterogeneous task deadlines	qlearning minus local
Heterogeneous task deadlines	qlearning minus local
Heterogeneous task deadlines	qlearning minus local
Heterogeneous task deadlines	qlearning minus local
Heterogeneous task deadlines	qlearning minus cloud
Heterogeneous task deadlines	qlearning minus cloud
Heterogeneous task deadlines	qlearning minus cloud
Heterogeneous task deadlines	qlearning minus cloud
Edge-1 compute-service outage/recovery	qlearning minus latency_greedy
Edge-1 compute-service outage/recovery	qlearning minus latency_greedy
Edge-1 compute-service outage/recovery	qlearning minus latency_greedy
Edge-1 compute-service outage/recovery	qlearning minus latency_greedy
Edge-1 compute-service outage/recovery	qlearning minus deadline_greedy
Edge-1 compute-service outage/recovery	qlearning minus deadline_greedy
Edge-1 compute-service outage/recovery	qlearning minus deadline_greedy
Edge-1 compute-service outage/recovery	qlearning minus deadline_greedy
Edge-1 compute-service outage/recovery	qlearning minus edge
Edge-1 compute-service outage/recovery	qlearning minus edge
Edge-1 compute-service outage/recovery	qlearning minus edge
Edge-1 compute-service outage/recovery	qlearning minus edge
Edge-1 compute-service outage/recovery	qlearning minus local
Edge-1 compute-service outage/recovery	qlearning minus local
Edge-1 compute-service outage/recovery	qlearning minus local
Edge-1 compute-service outage/recovery	qlearning minus local

Edge-1 compute-service outage/recovery	qlearning minus cloud
Edge-1 compute-service outage/recovery	qlearning minus cloud
Edge-1 compute-service outage/recovery	qlearning minus cloud
Edge-1 compute-service outage/recovery	qlearning minus cloud

Metric	Mean paired difference	95% CI low	95% CI high
mean_latency	2.736717	2.021304	3.452131
p95_latency	1.679217	-1.237096	4.595529
p99_latency	33.812599	8.721046	58.904152
deadline_violation_rate	0.000000	0.000000	0.000000
mean_latency	2.736717	2.021304	3.452131
p95_latency	1.679217	-1.237096	4.595529
p99_latency	33.812599	8.721046	58.904152
deadline_violation_rate	0.000000	0.000000	0.000000
mean_latency	-3.517690	-4.259024	-2.776356
p95_latency	-14.570003	-19.131736	-10.008271
p99_latency	12.582931	-9.799771	34.965632
deadline_violation_rate	-0.001370	-0.005173	0.002433
mean_latency	-8.091115	-8.811656	-7.370574
p95_latency	-49.215144	-60.484404	-37.945885
p99_latency	-44.395268	-62.761515	-26.029021
deadline_violation_rate	-0.014027	-0.025092	-0.002963
mean_latency	-76.271170	-77.251147	-75.291193
p95_latency	-104.064335	-107.566553	-100.562117
p99_latency	-58.514857	-83.962991	-33.066723
deadline_violation_rate	-0.880917	-0.885198	-0.876635
mean_latency	1.710527	1.327911	2.093143
p95_latency	0.231783	-1.268589	1.732155
p99_latency	16.841644	-8.962517	42.645804
deadline_violation_rate	0.000680	-0.001208	0.002569
mean_latency	1.710527	1.327911	2.093143
p95_latency	0.231783	-1.268589	1.732155
p99_latency	16.841644	-8.962517	42.645804
deadline_violation_rate	0.000680	-0.001208	0.002569
mean_latency	-2.266860	-2.837987	-1.695732
p95_latency	-6.460747	-8.666573	-4.254922
p99_latency	21.548040	13.439394	29.656686
deadline_violation_rate	0.001384	-0.000970	0.003739
mean_latency	-5.863223	-6.331518	-5.394927
p95_latency	-42.147751	-65.816175	-18.479328
p99_latency	-10.510850	-21.234880	0.213180
deadline_violation_rate	-0.005562	-0.010431	-0.000693

mean_latency	-74.046474	-75.701695	-72.391253
p95_latency	-95.770843	-98.452889	-93.088797
p99_latency	-26.915313	-38.568402	-15.262224
deadline_violation_rate	-0.880538	-0.895411	-0.865665

SD paired difference	Paired t p-value (exploratory)	Paired seeds
0.576173	0.000445	5
2.348713	0.185137	5
20.208005	0.020096	5
0.000000		5
0.576173	0.000445	5
2.348713	0.185137	5
20.208005	0.020096	5
0.000000		5
0.597049	0.000192	5
3.673886	0.000893	5
18.026376	0.193583	5
0.003063	0.373901	5
0.580303	0.000006	5
9.075933	0.000265	5
14.791640	0.002566	5
0.008911	0.024451	5
0.789245	0.000000	5
2.820585	0.000000	5
20.495185	0.003089	5
0.003448	0.000000	5
0.308148	0.000242	5
1.208356	0.690075	5
20.781919	0.144194	5
0.001521	0.373901	5
0.308148	0.000242	5
1.208356	0.690075	5
20.781919	0.144194	5
0.001521	0.373901	5
0.459970	0.000385	5
1.776508	0.001244	5
6.530467	0.001799	5
0.001896	0.177893	5
0.377151	0.000004	5
19.061859	0.007793	5
8.636821	0.052918	5
0.003921	0.033806	5

1.333067	0.000000	5
2.160042	0.000000	5
9.385058	0.003038	5
0.011978	0.000000	5

Earlier technical audit — archival context only

Audit snapshot	Seed	Policy	Generated	Completed	Pending
Member 3 earlier audit		1 Frozen RL	228	224	4
Member 3 earlier audit		1 Local-only	228	218	10
Member 3 earlier audit		1 Edge-only	228	222	6
Member 3 earlier audit		1 Cloud-only	228	183	45
Member 3 earlier audit		2 Frozen RL	228	222	6
Member 3 earlier audit		2 Local-only	228	215	13
Member 3 earlier audit		2 Edge-only	228	218	10
Member 3 earlier audit		2 Cloud-only	228	178	50

Completion rate	Average True E2E	Measurement cutoff	Deadline setting
98.25%	11.1952	500	Fixed global deadline 500
95.61%	17.4009	500	Fixed global deadline 500
97.37%	14.2531	500	Fixed global deadline 500
80.26%	81.6142	500	Fixed global deadline 500
97.37%	9.9644	500	Fixed global deadline 500
94.30%	17.2413	500	Fixed global deadline 500
95.61%	13.7110	500	Fixed global deadline 500
78.07%	82.2194	500	Fixed global deadline 500

	Status
	Legacy only; do not merge with v26
	Legacy only; do not merge with v26
	Legacy only; do not merge with v26
	Legacy only; do not merge with v26
	Legacy only; do not merge with v26
	Legacy only; do not merge with v26
	Legacy only; do not merge with v26
	Legacy only; do not merge with v26

Targeted research-gap matrix v2

ID	Database	Year
SD-01	ScienceDirect	2026
SD-02	ScienceDirect	2026
SD-03	ScienceDirect	2026
SD-04	ScienceDirect	2026
SD-05	ScienceDirect	2026
SD-06	ScienceDirect	2026
SD-07	ScienceDirect	2025
SPR-01	SpringerLink	2026
SPR-02	SpringerLink	2026

SPR-03	SpringerLink	2025
SPR-04	SpringerLink	2025
IEEE-01	IEEE Xplore	2026
IEEE-02	IEEE Xplore	2026
IEEE-03	IEEE Xplore (metadata found; secondary index used for DOI cross-check)	2026
PRIOR-01	IEEE Xplore / IEEE conference record	2023
PRIOR-02	IEEE ICC / user-provided paper record	2026
PRIOR-03	MDPI / supplementary prior art	2026
PRIOR-04	IIETA / supplementary prior art	2025
SPR-05	SpringerLink	2026

SPR-06	SpringerLink	2026
SD-08	ScienceDirect	2026

29 — 21 records; not systematic

Title	DOI
Forecast-driven task offloading for reliable and adaptive mobile edge computing	10.1016/j.jnca.2026.104494
RELTO: A reliability-oriented DRL approach with context-aware adaptive reward weighting for multi-objective task offloading in MEC	10.1016/j.adhoc.2025.104065
Chance constraint-based dependent task offloading for multi-access edge computing under uncertainty	10.1016/j.asoc.2026.115710
Deadline-Aware joint optimization of task offloading and resource allocation for cell-Free mobile edge computing networks	10.1016/j.comnet.2026.112219
Joint offloading and energy optimization in NOMA-aided UAV-MEC under practical impairments: A continuous contextual bandit DRL framework	10.1016/j.phycom.2026.103309
A fault-tolerant task offloading framework via large-scale multi-objective evolutionary optimization and game-based decision mechanism	10.1016/j.eswa.2025.129910
DELTA: Deadline aware energy and latency-optimized task offloading and resource allocation in GPU-enabled, PiM-enabled distributed heterogeneous MEC architecture	10.1016/j.sysarc.2025.103335
Multi-armed bandit-based federated reinforcement learning for dynamic task offloading in a containerized MEC	10.1007/s42452-026-08386-7
Dynamic task offloading strategy in mobile edge computing using meta-reinforcement learning with adaptive learning rate adjustment	10.1007/s44443-026-00736-0

Computation offloading based on incomplete information in edge computing networks	10.1007/s10586-025-05560-1
Time-and-Traffic-aware collaborative task offloading with service caching-replacement in cloud-assisted mobile edge computing	10.1007/s10586-025-05629-x
Task Offloading and Resource Optimization Based on Dependency-Aware Graph and Collaborative Deep Reinforcement Learning in Mobile Edge Computing	10.1109/TMC.2026.3665729
CVaR-Constrained Reinforcement Learning for Ultra-Reliable Task Offloading in LEO Satellite Edge Computing	10.1109/PRMVAI70103.2026.11605648
Risk-Aware Graph-Collaborative Deep Reinforcement Learning for UAV-Enabled MEC Task Offloading	10.1109/ICETCI68937.2026.11610771
Energy-Efficient Deadline-Aware Edge Computing (E2DA)	10.1109/GLOBECOM54140.2023.10437602
Chance-Constrained Task Offloading for Reliability Guarantees in Multi-Tenant Networks (O2C/CCMAB)	10.1109/ICC59461.2026.11586921
Lemur-Inspired Task Offloading for Edge–Fog–Cloud Continuum Systems (LITO)	10.3390/s26051497
QoS-Aware Task Offloading in Fog-IoT Using Hybrid Deep Q-Learning with Actor-Critic (H-DQAC)	10.18280/jesa.581016
Fuzzy chance constraint-based task offloading for MEC under uncertainties with joint multi-network collaboration and privacy preservation	10.1007/s44443-026-00567-z

Lyapunov Optimization with Virtual Queues for adaptive task offloading in multi access edge computing	10.1007/s10791-026-10660-6
Risk-aware user satisfaction maximization in vehicle-assisted multi-access edge computing offloading: a game-theoretic approach	10.1016/j.dcan.2025.11.001

Link

https://doi.org/10.1016/j.jnca.2026.104494

https://doi.org/10.1016/j.adhoc.2025.104065

https://doi.org/10.1016/j.asoc.2026.115710

https://doi.org/10.1016/j.comnet.2026.112219

https://doi.org/10.1016/j.phycom.2026.103309

https://doi.org/10.1016/j.eswa.2025.129910

https://doi.org/10.1016/j.sysarc.2025.103335

https://doi.org/10.1007/s42452-026-08386-7

https://doi.org/10.1007/s44443-026-00736-0

https://doi.org/10.1007/s10586-025-05560-1

https://doi.org/10.1007/s10586-025-05629-x

https://doi.org/10.1109/TMC.2026.3665729

https://doi.org/10.1109/PRMVAI70103.2026.11605648

https://doi.org/10.1109/ICETCI68937.2026.11610771

https://arxiv.org/html/2308.06647

https://hal.science/hal-05705287v1

https://www.mdpi.com/1424-8220/26/5/1497

https://iieta.org/journals/jesa/paper/10.18280/jesa.581016

https://link.springer.com/article/10.1007/s44443-026-00567-z

https://link.springer.com/article/10.1007/s10791-026-10660-6

https://www.sciencedirect.com/science/article/pii/S2352864825001750

Methodology

TimesNet multi-step forecast of server load and failure dynamics + LinUCB contextual-bandit offloading; real-world multivariate traces and simulated failure processes.

PPO with hybrid action space and dual-attention mechanism for context-aware adaptive weighting of reliability, latency, and energy objectives.

Uncertain joint optimization for dependent tasks; decomposed chance-constrained subproblems solved with improved genetic algorithm, Monte Carlo simulation, and ANN approximation.

JAB-TORA integrates BS–UE association, beam design, task split ratio, and resource allocation; MADRL is used for dynamic channel/power decisions.

Continuous contextual bandit Soft Actor-Critic (CCB-SAC), gamma=0, jointly optimizes UAV 3D position, power-allocation ratio, and task-offloading ratio using analytical probability boundaries as reward.

ALC-LSMOEA-KT large-scale multi-objective evolutionary optimization with knowledge transfer, plus Stackelberg game-based migration for recovery.

Multi-objective mixed-integer programming formulation and deadline-aware energy/latency offloading/resource allocation in a heterogeneous cooperative MEC architecture.

Hybrid MAB for lightweight device-layer real-time decisions plus federated reinforcement learning across edge servers in a containerized MEC environment.

ALR-MRL: meta-reinforcement learning with Seq2Seq task-graph embeddings and adaptive learning rates based on gradient variance/bias in a cloud–edge–end architecture.

Distributed auction multi-armed bandit (DA-MAB/TODAuction-MAB) plus heterogeneous distributed MAB (HD-MAB); sliding-window adaptation to non-stationary rewards and regret analysis.

Irregular Cellular Learning Automata classify traffic patterns/timescales; distributed deep RL for decentralized offloading; dynamic 0–1 knapsack for service caching; Q-learning for service replacement.

Dependency-aware graph feature extraction plus graph-based multi-agent DRL (DGMA), adaptive prioritized replay, and correlation-based selective parameter sharing.

CVaR-constrained risk-aware reinforcement learning for task offloading in a dynamic LEO satellite edge-computing system.

Graph-collaborative DRL with a distributional value critic and CVaR regularization; hybrid discrete/continuous offloading/resource actions.

Neural epsilon-greedy contextual multi-armed bandit; context includes task size, computational intensity, and deadline; action is local processing or offloading via a wireless channel.

Chance-constrained multi-tenant server assignment; KL-based upper-confidence estimates of feasible co-tenancy/success probability; integer linear program for allocation.

Hierarchical VFDT local/offload decision, contextual CSPL-CB support-node selection, EDF–DRS scheduling, feasibility filter and fallback.

Hybrid DQL/Double-DQN and Actor-Critic; local/fog/cloud actions; iFogSim2 evaluation.

Fuzzy optimization model transformed into fuzzy chance-constrained optimization with predefined confidence levels; hybrid improved adaptive genetic algorithm + Monte Carlo simulation + neural-network surrogate evaluation.

Lyapunov Drift-Plus-Penalty with virtual energy queues, decay-aware queue updates and battery-aware admission/offloading decisions.

Prospect Theory models user risk-aware behavior; offloading is formulated as a non-cooperative game with a proven unique Pure Nash Equilibrium and a low-complexity distributed iterative optimization algorithm.

Deadline / risk treatment
Reliability-aware and failure-aware using forecasted context; abstract does not establish a formal per-task chance constraint.
Reliability explicitly optimized alongside latency and energy; exact formal guarantee not verified from accessible record.
Explicit chance constraints under stochastic uncertainty; jointly addresses latency, energy, task utility, caching, multipath transmission, and privacy.
Objective minimizes a weighted sum of average delay and deadline-related dropout ratio.
Derives successful-computation probability and energy-outage probability under imperfect CSI/SIC; quasi-static environment is explicitly reformulated as a contextual bandit.
Jointly models delay, energy, load balance, and risk; fault-tolerant recovery in dynamic MEC.
Explicitly handles application deadlines and task dependencies while minimizing UE energy and application latency.
Focuses on dynamic workloads, QoE, latency and energy; no explicit formal chance constraint identified in the reviewed abstract.
Optimizes latency on synthetic DAG workflows and varying resources/bandwidth; no formal chance constraint established in the inspected article sections.

Latency-sensitive allocation subject to network latency and incomplete/local feedback; no explicit per-task probabilistic deadline guarantee in the abstracted model.

Time/traffic-aware latency and energy optimization; not a formal chance-constrained reliability method.

Deadline violation rate is an explicit reported metric; handles dependent tasks, dynamic network conditions and partial observability.

Explicitly constrains tail risk via CVaR to improve reliability beyond expected-latency optimization.

Explicit tail-risk suppression under stochastic traffic, channel blockage and energy constraints.

Deadline-aware reward uses an indicator for meeting the deadline; optimizes energy efficiency and response time.

Explicit constraint P(delay <= task latency limit) >= target reliability; derives admissibility and regret guarantees.

Heterogeneous deadlines, queue/resource feasibility, safe fallback, node deactivation and redistribution experiments.

Deadline-miss count is part of reward, alongside normalized latency and energy.

Explicit fuzzy chance constraints for uncertainty/privacy; jointly optimizes latency, energy, multi-network data allocation and privacy. Not a contextual-bandit method.

Hard task deadlines are part of the feasibility checks; primary focus is long-term energy constraints, queue stability and latency/energy trade-off, not a contextual bandit or formal latency chance constraint.

Models uncertain vehicle-provided compute resources and risk-aware user satisfaction; abstract reports lower vehicle failure probability. Not a contextual bandit and not a formal per-task deadline chance constraint.

Baselines / metrics evidenced
Publisher record reports lower task failures/system cost and faster convergence; full exact baseline table not extracted in this audit.
Publisher abstract/highlights report reliability, energy and latency comparisons; complete baseline table not extracted.
Publisher abstract reports comparison with five algorithms; exact names/table not extracted.
Publisher record describes comparisons with representative optimization- and learning-based benchmarks; exact names not fully extracted.
CHEA, EHGA, ABC, PSO and standard MDP-based DRL; successful computation probability, energy outage, optimality gap and inference complexity.
Publisher abstract describes multi-objective optimization and fault-tolerant comparisons; complete baseline table not extracted.
Publisher abstract lists Intelligent-TO, Multi-user, and Selective-random policies; latency and UE energy.
Publisher reports comparisons with local/greedy offloading and centralized DRL; QoE, latency, energy.
Greedy, fine-tuning DRL, MRLCO and HEFT-like priority scheduling; latency across resource/bandwidth groups.

Simulation studies of latency-sensitive rewards, average delay, resource utilization, server heterogeneity and regret.

Publisher abstract reports comparison against traditional centralized approaches; detailed baseline table should be extracted if this paper is retained as a core comparator.

IEEE metadata/abstract report delay, energy, offloading utility and deadline violation rate; exact baseline table requires authorized full text.

IEEE abstract states latency and reliability comparisons against baselines; exact algorithms and numerical tables not extracted.

Secondary bibliographic record describes the method; full official IEEE record and baseline tables not independently accessed.

Oracle-style EEL*, EE*, R*; response time and energy efficiency; ns-3/LTE/MEC simulation.

Reliability and regret under soft/strict settings; full PDF was previously inspected in project work.

Latency, SLA violations, task success, energy, utilization, throughput/fairness, ablations.

Q-learning, DQN, multi-agent DRL, federated DRL; latency, task success, deadline misses, energy, utilization, throughput/QoE.

Springer full text states benchmarking against four algorithms; reports at least 23.24% average objective-value reduction. Exact baseline names should be extracted from the comparison table if used as a core baseline.

Local execution, random offloading, edge-only, and classical Lyapunov-based offloading; reports average delay and energy.

Publisher abstract reports comparisons with benchmark methods; full baseline names and equations were not independently extracted in this pass.

Overlap with our project
Very high for contextual bandit + adaptive offloading + reliability/failure context; forecasting is not implemented in our project.
High for reliability-aware offloading and context-sensitive objective weighting; current project has a fixed weighted reward and no adaptive weights.
High for the broad claim of uncertainty-aware/chance-constrained task offloading, but the modeled problem differs substantially from independent local/edge/cloud actions.
High for deadline-aware offloading and deadline/dropout metrics; distinct cell-free wireless architecture and joint radio-resource decisions.
Very high methodological warning: contextual single-step learning with gamma=0 is explicitly used in a 2026 offloading paper. System setting differs (UAV/NOMA/continuous joint control).
Adjacent prior art for failure-aware/fault-tolerant offloading; algorithm family differs.
High for deadline-aware latency optimization, although it is optimization-based and includes GPU/SSD/PiM resources and dependent applications.
High for MAB-based dynamic offloading; adds federated coordination and containerized deployment absent from our project.
Adjacent for adaptive offloading under changing resources, three-tier execution, and task-graph state; more complex temporal/meta-learning method.

High for online bandit offloading under incomplete state and non-stationary network/server load.

High for Q-learning and cloud-assisted edge offloading under changing traffic; caching/replacement is outside our current scope.

High for RL offloading and deadline-violation evaluation; task dependency and multi-agent graph learning are not in our current model.

High for the broad claim that risk-aware/catastrophic-tail offloading is novel; application setting is satellite edge rather than iFogSim2 edge-cloud.

High for generic risk-aware DRL/CVaR claims, but UAV-specific action space differs.

Very high for contextual-bandit + deadline-aware reward, although the system/network and objective differ.

Very high for any proposed claim about probabilistic deadline reliability or chance-constrained offloading.

Very high for edge–fog–cloud, heterogeneous deadlines, contextual bandit, node failure and fallback.

Very high for same simulator family, action set, deadline-aware reward and RL offloading.

Relevant to uncertainty-aware and chance-constrained offloading, but its mathematical formulation and algorithm family differ from our tabular contextual policy.

Adjacent prior art for adaptive online offloading under dynamic resources, queues and deadlines; algorithmic family differs from our contextual-bandit configuration.

Relevant prior art for risk-aware offloading under uncertain compute capacity, but the mechanism is game-theoretic and vehicle-assisted.

Defensible distinction / implication

Only a narrow empirical distinction could be claimed if our contribution is specifically a reproducible iFogSim2 comparison of a simple tabular contextual policy against matched heuristics. Do not claim generic forecast-aware or reliability-aware bandit novelty.

A fixed tabular reward is a simpler baseline, not a new adaptive risk mechanism. Any new claim must identify a different mathematically specified mechanism and compare fairly.

Our current binary deadline-miss penalty is not a chance constraint. Do not claim probabilistic reliability unless a probability model, calibration, and constraint satisfaction evaluation are implemented.

Deadline-aware objective and violation/dropout reporting are not novel alone. Our distinct scope must be stated as an iFogSim2 edge–cloud empirical protocol, not a new deadline-aware principle.

The fact that gamma=0 can be appropriate in a quasi-static contextual problem is not novel. We must justify whether our sequential simulator really has no delayed consequences; otherwise use gamma>0 and test temporal credit assignment.

A generic claim of handling failures is not novel. If testing failure scenarios, define failure semantics and report them as comparative robustness experiments.

Deadline-aware task offloading is established. A defensible distinction must be in the exact experimental protocol or a genuinely different algorithm, not deadline inclusion alone.

A bandit for dynamic task offloading is already prior art. Our only plausible difference is a tightly scoped, reproducible iFogSim2 comparison under matched baselines and task semantics.

Resource-shift testing is useful but not novel by itself. If our task model has no DAG dependencies, that is a scope difference, not an algorithmic innovation.

Bandit adaptation to dynamic latency is already established. A claim must not be based only on discretized context or a changing workload.

Q-learning and traffic-aware cloud-edge offloading are established. Our possible distinction is a smaller controlled benchmark, not a new Q-learning/offloading concept.

Deadline-miss reporting and RL offloading are not novel. Our setting is simpler; comparison should acknowledge the different task-dependency/agent architecture.

Do not claim CVaR/tail-risk-aware offloading as a new general idea. Our current model does not implement CVaR at all.

Keep as a supplementary prior-art alert with evidence grade limited; do not rely on it for detailed equations until official full record/text is obtained.

A deadline feature and miss penalty in a contextual bandit cannot be claimed as novel alone.

Current binary reward penalty is not a chance constraint and offers no probabilistic guarantee; a new formulation would need a distinct constraint/model and direct comparison.

Deadline classes and outage tests alone are not novel; LITO should be in the mandatory equation/experiment comparison.

Using iFogSim2 or Q-learning with a deadline penalty is not novel by itself.

Current project does not implement fuzzy chance constraints or privacy constraints. This paper is evidence that uncertainty-aware offloading is prior art, not that the same problem is solved.

Dynamic offloading and queue-aware adaptation are not new alone. Our comparison should specify the learning policy and fair experimental protocol rather than claim general adaptive offloading novelty.

Risk-aware task offloading is established broadly. Our current binary deadline penalty does not implement Prospect Theory or a calibrated risk model.

Evidence grade	Publisher evidence
Publisher full page/abstract and indexed algorithm text reviewed; exact full equations/tables not independently transcribed.	Official ScienceDirect page
Publisher abstract/highlights and indexed text; full equations/tables not independently extracted.	Official ScienceDirect page
Official publisher abstract and section snippets; full equations/table not independently extracted.	Official ScienceDirect page
Official ScienceDirect abstract/highlights and indexed sections.	Official ScienceDirect page
Official ScienceDirect full-length page/indexed full text reviewed; DOI and abstract verified.	Official ScienceDirect page
Official ScienceDirect abstract/highlights.	Official ScienceDirect page
Official ScienceDirect abstract.	Official ScienceDirect page
Official SpringerLink open-access full text/abstract reviewed.	Official SpringerLink page
Official SpringerLink full text reviewed.	Official SpringerLink page

Official SpringerLink open-access full text reviewed; system model and regret discussion inspected.	Official SpringerLink page
Official SpringerLink open-access full text/abstract reviewed.	Official SpringerLink page
Official IEEE Xplore metadata/abstract; full paper not available in this audit.	Official IEEE Xplore page
Official IEEE Xplore abstract/metadata; full paper not extracted.	Official IEEE Xplore page
Secondary bibliographic record/abstract only; official full text not verified.	DOI metadata cross-checked using secondary index
Full preprint text reviewed.	Open full-text preprint; DOI verified
User-provided full PDF previously reviewed; HAL record/link available.	
Full text/indexed equations and algorithms reviewed in previous audit.	
Official SpringerLink full text reviewed; system model and methodology sections accessible.	Official SpringerLink full text

Official SpringerLink full text reviewed.	Official SpringerLink full text
Official ScienceDirect abstract and metadata reviewed; full equation/table extraction not completed.	Official ScienceDirect record

Supplied/core paper review matrix

literature comparis	
8 supplied/relevant records counted once; the 9th record is an	
ID	Paper
P1	RL-Based Adaptive Task-Offloading in Mobile-Edge Computing for IoT Networks
P2	Edge-Cloud Adaptive Framework for Real Time IoT Data Processing in Smart Cities
P3	Deep Q-Network based Task Offloading Optimisation for Edge-IIoT System
P4	Deep Learning-Centric Task Offloading in IoT-Fog-Cloud Continuum: State-of-the-Art Review
P5	An Adaptive Edge-Cloud Task Offloading Framework using RL for Latency-Sensitive AI Applications

P6	An Analysis of Task Offloading Approaches in Edge-Cloud Continuum
P7	Hierarchical DRL-Based Partial Task Offloading in Device-Edge-Cloud Architecture
P8	Task Offloading for CAVs Edge Computing Environment: Taxonomy, Critical Review, and Future Road Map
P9	An efficient computational offloading method using deep reinforcement learning in edge-end-cloud

Venue/year	DOI / citation key
IEEE Internet of Things Magazine, 2025	DOI 10.1109/MIOT.2025.3623451
IEEE ACROSET, 2025	DOI 10.1109/ACROSET66531.2025.11280634
IEEE ICICIT, 2026	DOI 10.1109/ICICIT69063.2026.11634047
IEEE Access, 2025	DOI 10.1109/ACCESS.2025.3599190
IEEE ICOECA, 2026	DOI 10.1109/ICOECA68095.2026.11485334

IEEE COMPSAC, 2025 DOI 10.1109/COMPSAC65507.2025.00206

IEEE Transactions on Consumer Electronics, 2026 DOI 10.1109/TCE.2025.3611973

ACM Computing Surveys, 2026 DOI 10.1145/3783984 (ACM Computing Surveys 58(8), Article 194, Feb. 2026)

Ad Hoc Networks / Elsevier, 2025 DOI 10.1016/j.adhoc.2025.103941

Dataset / workload evidence	Simulator / testbed
No public workload trace identified. Simulated LTE/NOMA ultra-dense MEC; approx. 10–100 active nodes, up to 10 tasks/node, 5 edge servers; model-generated task/network conditions.	Simulation; modeled wireless MEC
102 traffic-camera streams described as CityScape-Traffic replayed for 4 rush-hour hours; 64 air-quality sensors at 1 Hz; synthetic bursts triple camera arrivals for 120 s every 30 min. Public data link/licence unverified.	Jetson Nano + AWS EKS; MQTT; WAN emulator
Mentions Kaggle/Edge-IIoT network data for telemetry/features. Separate fixed parameters include 100-KB task size, 500 mega-cycles, local CPU 1.5 GHz, edge 50 GHz, cloud 100 GHz, bandwidth 50 Mbps, transmit power 2 W. Network dataset ≠ complete scheduling benchmark.	Simulation (implementation details need independent reproduction)
Survey: no single dataset/workload applies to all reviewed papers.	Survey, not a single simulator
Parameterized/synthetic profiles: vision 8–20 MB, 3–8 Gcycles, 4–10 tasks/s; AR/VR 2–10 MB, 2–6 Gcycles, 6–14 tasks/s; health-IoT 0.5–3 MB, 1–4 Gcycles, 3–8 tasks/s; industrial CPS 1–6 MB, 2–7 Gcycles, 5–12 tasks/s. No public trace identified.	Simulated edge/cloud framework

EdgeCloudSim sample_app4, configured/generated app profiles (AR, health, heavy compute, infotainment); 200–1000 devices moderate, 2000+ high load; 5 configurations, 10 iterations. No public task trace identified.	EdgeCloudSim
Synthetic tasks (data size, CPU cycles, tolerable latency sampled in given ranges); Network I/II; no public task trace identified.	Authors' Python/PyTorch simulation; not stated to be iFogSim2/EdgeCloudSim
CAV task-offloading survey; discusses KITTI, Waymo Open, BDD100K and ApolloScape as automotive perception datasets. These are not task CPU/deadline/network outcome traces aligned for an iFogSim offloading benchmark.	Survey/taxonomy, not one experiment
Full-text PDF not provided; dataset/workload not verified from page/abstract.	Publisher abstract/page only; full text not audited.

State / action / method

Abstract/contribution says multi-agent DQN, but implementation section describes tabular Q-learning/Q-table; action relates to local vs edge/base-station assignment.

Double Dueling DQN; per-micro-batch Edge or Cloud; state includes cloud latency, backhaul bandwidth, CPU/GPU use, moving average latency.

DQN actions Local/Edge/Cloud; features include task size/cycles/deadline, latency/energy/bandwidth/server load.

No single state/action/reward. Provides taxonomy of DL/RL offloading approaches.

RL state includes edge utilization/queue, network rate, cloud queue and task features; action Edge/Cloud only.

Rule-based comparison, not learned policy; placement Local/Edge/Cloud depending on policy.

Hierarchical: high-level Double DQN chooses tier/actions; low-level PPO chooses split ratios across device, edge and cloud; fractions sum to one.

Not applicable.

Page describes threshold pre-stage, DQN + Q-learning, blockchain / integrity aspects in edge-end-cloud.





Reward / objective

Reward based on negative execution time; problem framing discusses energy and QoS delay constraints.

Normalized latency, bandwidth, energy (reward equation formatting in PDF is imperfect).

Reward-based objective reports latency, energy, cost, deadline success and utilization; exact reproduction needs equation/implementation audit.

Not applicable at survey level.

Penalty for latency plus deadline-met indicator.

Policy-specific heuristic objectives; not one common RL reward.

Multi-objective energy, task acceptance and resource use subject to latency constraints.

Not applicable.

Full reward/objective not verified from full text.





Baselines

Local, least-load / optimized variants (as described)

Cloud-only; static fog

Local-only, edge-only, cloud-only / offloading variants described in the paper.

Not applicable as one benchmark; paper surveys prior methods.

Threshold, greedy latency-minimization, Lyapunov/LODCO-like method.

NETWORK_BASED, UTILIZATION_BASED, FUZZY_BASED, FUZZY_COMPETITOR, HYBRID.

DRL-PTO (fixed all-tier split) and DRL-TO (whole-task placement).

Not applicable.

Not fully verified.





Metrics / reported result scope

Delay, energy, QoS/reliability, throughput-related outcomes.

Mean/P95 latency; 100-ms SLO satisfaction; backhaul; energy/frame; Jain fairness; overhead.

Latency, energy, cost, reward, utilization, on-time completion.

Maps metrics, methods, simulator/testbed and future research.

Mean/P95 latency, deadline success, queue/resource outcomes, energy (reported).

Failure ratio, service time, placement distribution.

Reward; acceptance ratio; energy; generalization across 100 runs/new task profiles.

Reviews workload, mobility, dataset, simulator and evaluation taxonomies.

Publisher page says approximately 40% reduction against strategies; exact conditions and baselines not independently audited.

Explicit author limits / future work

Explicit future directions: multi-objective latency/throughput/energy; joint computation/communication/caching; deep multi-agent RL; robustness/scaling.

Explicit future: policy-gradient alternatives, adaptive pruning; extra cloud GPU overhead for federated update is reported.

No crisp formal limitations section found; general Edge-IIoT challenges discussed in paper.

Survey discusses data scarcity, testbed/simulator limitations, energy, security, workload prediction and provisioning.

Explicit future: trust modeling, privacy-conscious offloading.

Future work: combine fuzzy adaptability and network awareness.

Future: Transformer/quantum DRL, meta-learning, digital twins.

Discusses real-time reliability/latency, energy, mobility, collaboration, discovery and generalization.

Full-text limitations not verified.





INFERRED limitations / open checks

INFERRED: DQN vs Q-table inconsistency harms reproducibility; simplified base-station-to-edge delay; no identified public workload trace or held-out topology test.

INFERRED: few baselines; 4K in abstract vs 1920x1080 in dataset section; month-long training trace vs 4-hour replay requires clarification.

INFERRED: task workload generation/reward details under-specified; network telemetry and task CPU/deadline fields mixed conceptually; no held-out topology/confidence-interval evidence clear.

INFERRED: survey-level gaps do not prove a specific method combination is unpublished; primary-paper comparison remains necessary.

INFERRED: no device/local action, waiting or partial splitting; data source appears to be generated ranges, not public trace.

INFERRED: no DQN/RL comparator or external trace identified; results tied to configured sample_app4 workload.

INFERRED: only 2 network configurations, no external trace, limited classical/optimization baselines; two-agent complexity significantly higher.

INFERRED: automotive mobility data should not be imported into stationary IoT study without an explicit motivation.

Classify as page/abstract-level evidence only; obtain PDF before making finer claims.

Adjacent prior-art scan — support

External overlap / novelty risk scan

Search cut-off 27 Oct 2026. These are close precedents, not an exhaustive proof of absence. Directly related papers should be checked in full text before manuscript submission.

Dimension
Latest-start-time / timing-dependent scheduling
Partial offloading + resource allocation
Partial offload + deadlines
Forecast + contextual bandit
Bandit + federated RL
Contextual bandit + deadline feasibility gate
Deadline/SLA-aware DRL

Guarded DRL

Chance constraints + uncertainty

Meta-RL / adaptation

Generalizable Pareto offloading

Dependency-aware offloading + cluster trace

Edge-end-cloud DRL + thresholding

Workload and trace data gap

ting evidence, not exhaustive

Title	Venue/year
Optimization of Task Scheduling Strategy With Timing and Data Dependencies in VEC Based on DRL	IEEE TMC, 2026
Joint optimization of multi-dimensional resource allocation and task offloading for QoE enhancement in Cloud-Edge-End collaboration	Future Generation Computer Systems, 2024
Delay-aware partial task offloading using multicriteria decision model in IoT–fog–cloud networks	JNCA, 2025
Forecast-driven task offloading for reliable and adaptive MEC	JNCA, 2026
Multi-armed bandit-based federated RL for dynamic task offloading in containerized MEC	Discover Applied Sciences, 2026
LITO: Lemur-Inspired Task Offloading for Edge–Fog–Cloud Continuum Systems	Sensors, 2026
SLA-aware deep reinforcement learning for adaptive EdgeCloud task scheduling	Scientific Reports, 2026

Guarded deep reinforcement learning for SLA latency and energy aware task scheduling in IoT edge cloud environments	Discover Computing, 2026
Chance constraint-based dependent task offloading for MEC under uncertainty	Applied Soft Computing, 2026
Dynamic task offloading strategy in MEC using meta-RL with adaptive learning rate	Journal of King Saud University C&IS, 2026
Generalizable Pareto-Optimal Offloading With RL in MEC	IEEE TSC, 2025
Dynamic Task Offloading in Edge Computing Based on Dependency-Aware RL	IEEE TCC, 2024
An efficient computational offloading method using DRL in edge-end-cloud	Ad Hoc Networks, 2025
Computation offloading in the edge-to-cloud compute continuum: a survey of federated architectural solutions	Cluster Computing, 2025





DOI

10.1109/TMC.2025.3646450

10.1016/j.future.2024.01.025

10.1016/j.jnca.2025.104278

10.1016/j.jnca.2026.104494

10.1007/s42452-026-08386-7

10.3390/s26051497

10.1038/s41598-026-40237-8

10.1007/s10791-026-10607-x

10.1016/j.asoc.2026.115710

10.1007/s44443-026-00736-0

10.1109/TSC.2025.3604371

10.1109/TCC.2024.3381646

10.1016/j.adhoc.2025.103941

10.1007/s10586-025-05577-6





Relevant overlap found

DAG timing/data dependencies; Latest Start Time metric; MADDPG

Dependent tasks, resource allocation, latency/energy objective; MAPPO.

Task partitioning, processing-efficiency criteria, deadlines and completion-time assignment.

TimesNet forecast + LinUCB; real-world multivariate traces + simulated failures.

Device MAB + federated PPO/DRL at edge; Local/Edge/Cloud; deadlines.

Contextual-bandit feedback + explicit capacity/deadline filtering and fallback.

SLA/deadline risk used in DRL scheduling.

Guarded DRL for SLA/latency/energy tradeoffs.

Dependent task offloading and probabilistic constraints under uncertainty.

Synthetic DAGs; meta-RL/Seq2Seq; adaptation across task complexity.

Multiobjective RL, varying preferences and MEC configurations.

DAG/dependency-aware RL; Alibaba cluster trace plus synthetic workload.

Threshold pre-stage, DQN/Q-learning; page-level evidence only.

Survey notes lack of datasets that align task, device, deadline, energy and network dimensions.





Consequence for our claim

Does not support novelty for ‘when to start’ alone.

Partial / multi-site task splitting is already studied.

No claim that deadline-aware partial offload is new.

Forecast + LinUCB is not a fresh novelty by itself.

Bandit, hybrid MAB/RL, and three-tier placement already overlap.

Very close precedent to naive bandit+deadline gate; exact algorithm and evaluation must be compared.

Deadline-aware deep scheduling not new.

Generic safe/guarded SLA scheduling has precedents.

Probabilistic deadline feasibility is not new in general.

Generalization alone is not a novelty claim; specify shifts and transfer protocol.

Generic multiobjective generalization already exists.

DAG scheduling and trace-based benchmarking already exist.

Edge-end-cloud DRL plus thresholding already exists.

Supports disclosed synthetic benchmark, not a claim of dataset novelty.





Evidence confidence / note

IEEE Xplore search record / DOI verified

ScienceDirect publisher record

ScienceDirect page says no data were used for the study.

ScienceDirect publisher page

Springer page; data-availability wording should be scrutinized.

MDPI publisher page, adjacent publisher used for overlap check.

Nature publisher page

Springer publisher page

ScienceDirect page

Springer page

IEEE Xplore record

IEEE Xplore record

Full PDF was not reviewed; avoid detailed limitation claims.

Springer publisher page

Combined search log — counts/exclusions still incomplete

Source log	Date	Database/source
v26 targeted search	2026-10-28	IEEE/Institutional record
v26 targeted search	2026-10-28	ScienceDirect
v26 targeted search	2026-10-28	MDPI Sensors full text
v26 targeted search	2026-10-28	IIETA publisher full text
v26 targeted search	2026-10-28	ScienceDirect
v26 targeted search	2026-10-28	ScienceDirect
v26 targeted search	2026-10-28	ScienceDirect
v26 targeted search	2026-10-28	ScienceDirect / institutional record
v26 targeted search	2026-10-28	arXiv
v29 targeted search	2026-10-28	SpringerLink
v29 targeted search	2026-10-28	SpringerLink

v29 targeted search	2026-10-28	ScienceDirect
v29 targeted search	2026-10-28	Count / quality correction

Query
"Chance-Constrained Task Offloading for Reliability Guarantees in Multi-Tenant Networks"
"Forecast-driven task offloading for reliable and adaptive mobile edge computing"
"LITO" "heterogeneous deadlines" "contextual bandit" edge fog cloud
"QoS-Aware Task Offloading" "Hybrid Deep Q-Learning" iFogSim2
"RELTO" reliability-oriented DRL context-aware adaptive reward weighting MEC
"risk-regularised DNN-PPO" edge fog strict SLA CVaR
"Chance constraint-based dependent task offloading" MEC uncertainty
"Deadline-Aware joint optimization" cell-Free MEC 2026
"Energy-Efficient Deadline-Aware Edge Computing" E2DA contextual neural bandit
Fuzzy chance constraint-based task offloading for MEC under uncertainties with joint multi-network collaboration and privacy preservation
Lyapunov Optimization with Virtual Queues for adaptive task offloading in multi access edge computing

Risk-aware user satisfaction maximization in vehicle-assisted multi-access edge computing offloading: a game-theoretic approach

Audit of v28 Literature Matrix and report statements

	Purpose
Verify O2C/CCMAB method, venue, DOI	
Verify recent forecast+bandit prior art	
Check overlap for deadline heterogeneity and node failure	
Check same-simulator RL prior art	
Check reliability-aware reward overlap	
Check tail-risk and SLA overlap	
Check chance-constrained optimization overlap	
Check deadline-aware RL and deadline-drop metrics	
Verify earlier direct prior art	
Official open-access full text: fuzzy chance constraints, multi-network allocation, privacy constraint, adaptive genetic algorithm + Monte Carlo + neural-network proxy; four baselines stated in article text.	
Official open-access full text: Lyapunov Drift-Plus-Penalty, virtual energy queues, decay-aware updates, battery-aware admission, local/edge decisions, four baselines.	

Official publisher abstract/metadata: Prospect Theory risk-aware behavior, non-cooperative game, unique Pure Nash Equilibrium, uncertain vehicle compute resources.

The v28 matrix contains 18 records: 14 additions (7 ScienceDirect, 4 SpringerLink, 3 IEEE records) plus 4 carried-forward anchors.

	Result / evidence
	Institutional record and DOI confirmed; abstract verified; full IEEE equations not extracted
	JNCA 2026, TimesNet + LinUCB, load/failure forecasting; abstract verified
	Full article reviewed; EDF-DRS, contextual bandit, feasibility filter, fallback, failure experiments are reported
	H-DQAC 2025, hybrid DQL/Actor-Critic, QoS metrics, iFogSim2; full publisher page reviewed
	PPO hybrid actions; reliability/latency/energy objective; publisher abstract verified
	CVaR regularization, SLA critic, p95/p99 evaluation; publisher page reviewed
	2026 Applied Soft Computing; chance-constrained dependent-task offloading
	Computer Networks 2026, DOI verified
	Full preprint and IEEE GLOBECOM DOI metadata available
	Adds direct evidence that uncertainty/chance-constrained offloading is prior art, though problem and method differ from our contextual policy.
	Adjacent prior art for online queue-aware energy-delay offloading; supports avoiding broad claims about adaptive offloading.

Adds prior-art evidence for risk-aware offloading under uncertain capacity.

Corrected prior report statement that said 13 additions plus 5 anchors; v29 matrix contains 21 records after adding 3 newly reviewed papers.

URL

https://researchportal.ip-paris.fr/fr/publications/chance-constrained-task-offloading-for-reliability-guarantees-in-/

https://doi.org/10.1016/j.jnca.2026.104494

https://www.mdpi.com/1424-8220/26/5/1497

https://iieta.org/journals/jesa/paper/10.18280/jesa.581016

https://www.sciencedirect.com/science/article/pii/S1570870525003130

https://www.sciencedirect.com/science/article/pii/S0140366426002008

https://www.sciencedirect.com/science/article/pii/S1568494626011580

https://www.sciencedirect.com/science/article/pii/S1389128626002318

https://arxiv.org/abs/2308.06647

Need inspect the baseline comparison table before quoting baseline names.

Different algorithm family; not a contextual-bandit paper.

Full equations and baseline names were not extracted in this pass.

The database search remains targeted, not exhaustive native-interface export.

Execution plan and closure gates

Priority	Workstream
P0	Supervisor decision
P0	Version/provenance
P0	Algorithm naming
P1	Native database evidence
P1	Full-text equation audit
P1	Task-level logging
P1	Metric definitions
P1	Replace proxies

P1	Baseline cleaning
P1	Confirmatory stats
P1	Generalization
P2	New risk method if required
P2	Dataset realism

Specific actions

Choose comparative/reproducibility study vs genuinely new algorithm. Confirm primary simulator and expected contribution level.

Tag/freeze v26 code, model artifact, config, topology, training seeds and held-out seeds; hash the release and preserve raw JSONL.

If γ=0 / contextual_bandit config remains, call it tabular contextual-bandit/Q-value learner. If sequential Q-learning is desired, implement γ>0 and test delayed consequences.

Export searches from IEEE Xplore, SpringerLink and ScienceDirect; save query, date, result count, export and screening columns. Record duplicate handling and exclusion reason.

Prioritize E2DA, O2C/CCMAB, LITO, H-DQAC, forecast-driven LinUCB, RELTO, CVaR and other closest papers. Mark equation/baseline inaccessible if paywalled.

Capture task_id, arrival, decision, action, chosen site, transfer start/end per hop, queue start/end, actual CPU start/end, first-module finish, full-loop finish, deadline, status/reason.

Define deadline relative to arrival; absolute deadline=arrival+relative deadline; fix late/failed/pending/censored denominator rules; never drop pending silently.

Implement actual CPU utilization and per-hop bytes/transfer times. Add energy only if model, units and accounting are validated.

Run Local/Edge/Cloud/Random/latency-greedy/current policy; remove deadline-greedy duplicate or redesign as truly distinct. Add DQN only if justified.

Freeze policies and tuning; run ≥20 paired seeds (target 30 if feasible), same task generation within seed, fresh held-out test seeds, report mean/P95/P99/CI and task-type strata.

Evaluate at least 2–3 topologies or carefully justified workload/network trace proxies; test load, WAN delay/bandwidth, edge-capacity and outage shifts separately first.

Specify residual quantile or other risk-calibration mechanism mathematically; distinguish it from LITO feasibility filtering and O2C chance constraints; tune only on validation.

Use documented synthetic generator as primary; optional Alibaba Cluster Trace for workload-shape proxy, Edge-IIoTset for network telemetry proxy, Topo4MEC for topology only.

Owner	Deliverable/evidence
Supervisor + Mark + team	Written decision and agreed research question
Simulator owner + Mark	Version sheet + manifest + code/model/config identifiers
Model owner + supervisor	Algorithm description matching config and code
Research Gap owner	Native exports + screening log
Research Gap owner + Member 1	Claim-by-claim equation and baseline matrix
Simulator owner	Schema + validated raw logs
Simulator owner + supervisor	Metric specification sheet
Simulator owner	Telemetry and unit tests

Algorithm owner	Pinned baseline implementations and configs
Whole team	Final result CSV + test protocol
Whole team	Scenario matrix + manifests
Algorithm owner + research owner	Algorithm equations + ablation + baseline protocol
Data owner	Generator/config/data mapping and license notes

Exit criterion

No new method built until this is confirmed.

Every result row traces to one exact configuration.

Paper, report and code use consistent name.

Search trail is reproducible; no invented result counts.

Each novelty distinction linked to text/equation or labelled uncertain.

One terminal status per task; no mismatches/orphans.

Every metric reproducible from logs; units validated.

No CPU/network/energy claim based solely on proxy.

Same workload/seed, same simulator and metric definitions.

No tuning on final test; report variability and confidence intervals.

Repeatable tests with isolated factors and explicit limits.

Evidence of substantive difference and held-out benefit, else reject novelty claim.

No source is mislabelled as complete task-offloading data.

	Dependency/status
	Immediate / open
	Immediate / v26 main; v25 and Seed1/2 separate
	Immediate
	After meeting; focused v29 audit done
	Some source evidence only abstract-level
	Needs code instrumentation
	Before formal experiments
	Before resource/energy claims

Before confirmatory runs

After metric and policy freeze

After base correctness

Only if supervisor requires a new method

After primary benchmark is sound

Practical requirements extracted from papers + project audit

Requirement	Derived from literature / experiments
Per-task task model	Deadline-aware / QoS papers; current v26 scenario
Context state	DQN/Bandit/forecast-driven papers
Action space	Local/Edge/Cloud papers
Objective / reward	E2DA, O2C, LITO, H-DQAC, RELTO, CVaR studies
Baselines	Static baselines and strong deterministic heuristics in supplied works
Latency/Deadline metrics	Vashisth, Swapna, H-DQAC; current output
Resource and network accounting	Papers using throughput, bandwidth and utilization
Failure semantics	LITO, forecast-driven MEC, outage test
Robustness / uncertainty	O2C, forecast-LinUCB, CVaR and workload-shift studies

Data provenance	Continuum survey and data audit
Reproducibility	All empirical comparisons
Research-gap evidence	21-row targeted source matrix

	Current state
	MI/data class present; v26 uses 20/50/120 deadline profile
	MI, WAN latency, backlog estimate; utilization remains proxy
	Three actions; Edge target may be parent edge not separate Edge-1/2 action
	γ=0 contextual-bandit config; reward/latency/deadline flags need frozen definition
	Local/Edge/Cloud/latency-greedy; Q policy; duplicate deadline-greedy; random in other configs
	Sensor-to-first-module completion; deadline violations and P95/P99 available
	Backlog proxy, payload bytes proxy; per-hop total not complete
	Edge-1 service masking for new assignments; router active/in-flight tasks continue
	Several synthetic scenarios; v26 only five test seeds per scenario-policy and one topology

No fully aligned public dataset verified

Code and raw logs bundled; hashes/commit and native-search exports still need final consolidation

No broad algorithmic gap confirmed; evidence grade varies

Required action
Log task arrival, MI, input bytes, class, relative+absolute deadline and seed/profile.
Verify feature units/ranges and add queue/resource signals only if measured.
Document action mask and actual target; decide if edge nodes should be separate actions.
State objective explicitly; no chance guarantee unless modelled.
Freeze baseline roster and code; avoid counting duplicate implementations separately.
Add full-loop completion and per-task outcome; ensure all units consistent.
Instrument CPU start/finish, queue time, link transfer/bytes on each hop.
Keep compute-service outage distinct from host crash/network failure.
Use paired fresh seeds, 20 minimum / target 30, 2–3 topologies if feasible.

Document synthetic generator; optional partial proxies separately.

Store model, config, seed list, run manifest, raw logs, code release ID.

Complete publisher records + equation-level check; final search exports and exclusions.

Acceptance criterion
Deadline does not depend on selected site/policy; one task ledger row per generated task.
State values reconstructable from logged telemetry; no train-test leakage.
Action in policy log matches executed site; zero routing mismatch.
Reward/constraints mathematically specified, unit tested, and not changed between baselines.
Same scenario, seed, workload, cutoff, drain and metrics for every policy.
Metric names match exactly what is measured; pending/censored is explicit.
No utilization or total-traffic claim based on proxy.
Scenario definition includes what happens to in-flight tasks and recovery.
Report mean and CI, tails, misses and stratified results.

Every task input is traceable to generator/trace mapping, version, seed and assumptions.

Independent team member can reproduce one seed and regenerate summary.

Final claim is narrow, sourced and approved by supervisor.

Dataset/workload option audit

Dataset/workload source audit

No single public dataset verified as containing all required task, deadline, network, resource and outcome fields in one aligned record. Public sources can inform one component at a time.

Candidate source

Alibaba Cluster Trace Program, v2017/v2018

Edge-IIoTset

Topo4MEC

iFogSim2

Cloud-Edge Collaborative Task Scheduling GitHub repo

Main generated workload (recommended)

What it actually contains

Real production cluster traces; v2017 ~1300 machines / 12 h; v2018 ~4000 machines / 8 days and task DAGs.

Public IIoT/cybersecurity/network telemetry dataset; source paper DOI 10.1109/ACCESS.2022.3165809.

Topology edge-list files and ingress nodes; includes random graphs and Milan City Center topology derived from OpenCellID.

Simulator toolkit; repository states mobility support, migration, microservice orchestration and dynamic distributed clustering.

README describes Topo4MEC/Pakistan topology use and synthetic task attributes/deadlines; no license verified in scan.

Generate arrivals, bytes, MI/cycles, class and relative deadline from a documented configuration; at least steady Poisson, bursty/on-off, mixed task/deadline profiles.





Useful role

Workload / DAG-arrival-shape proxy, not full IoT edge data.

Optional network-event / telemetry proxy only, with clear mapping.

Optional topology stress/generalization after core benchmark works.

Execution/simulation platform.

Possible inspiration / audit candidate only.

Primary benchmark because it can be aligned exactly with iFogSim2 task/event logging.





What it does not provide

Does not inherently provide per-task relative deadlines, network delays, IoT input sizes and iFogSim2 execution outcomes aligned in one record.

Not a native task-offloading benchmark: cannot assume CPU cycles, per-task deadlines or site-specific execution outcomes come from it.

Topology only, not task-arrival/deadline/compute workload.

Not a dataset and doesn't remove need to verify event timing and resource accounting.

Do not rely on it until license, data schema, generator, commit and runtime are checked.

Synthetic, therefore no claim of real-world representativeness without calibration and sensitivity analysis.

License / usability checks

Data downloads may require a brief survey; use research/study purposes and follow the trace docs. Check dataset version, citations and policy before using.

Review dataset terms and license on the official page; do not relabel network records as compute jobs.

Repository README states GPL-3.0; verify license/attribution requirements and whether its abstraction fits the current iFogSim topology.

Pin the git commit and configuration for reproducibility.

If adopted, write a small independent validation and preserve attribution; not a validated primary dataset yet.

Use fixed seeds, publish the generator/configuration; choose task deadlines independently of the chosen policy/site; include ranges from the reviewed papers as scenario settings, not as validated ground truth.

Source URL

https://github.com/alibaba/clusterdata

http://ieee-dataport.org/8939

https://github.com/bnxng/Topo4MEC

https://github.com/Cloudslab/iFogSim

https://github.com/henrenzhenderen-wangyu/Cloud-Edge-Collaborative-Task-Scheduling

Proposed, not an external dataset.